

# **Carbon Capture Proposal and Technical Assessment for DTE's Blue Water Energy Center**

ESENG 503 Project Report

Submitted to:

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## Executive Summary

In this analysis, a carbon capture system is designed and analyzed for efficacy in capturing 95% of the total carbon emissions at DTE's Blue Water Energy Center natural gas combined cycle power plant at 95% CO<sub>2</sub> purity. This is done primarily to achieve lower emissions for Blue Water via carbon capture and sequestration. I found that a system could be sufficiently sized to handle an average 2,624,743.32 metric tons of CO<sub>2</sub> per year (2,893,280.81 US tons per year) up to 3,335,409.15 metric tons of CO<sub>2</sub> per year (3,676,354.86 US tons per year) at maximum emission output for Blue Water. The cost of the system will be \$285,610,727.44, constituting two Absorber Columns, a single Regeneration Column, a CO<sub>2</sub> Compressor, a new Cooling Tower, Solvent Heat Exchanger, Reboiler Heat Exchanger, Flue Gas Blower, and several major condensers and pumps. The auxiliary power demand of the equipment will be 64.56 MW, and the new cooling load will be 409,178.11 kW<sub>thermal</sub> requiring 186,128.94 gpm of cooling water from the St. Clair River.

The most significant change to the plant's operation will be the total derate in maximum gross power, from 1,127 MW to 825.77 MW. This change is due to the 301.23 MW decrease in steam power due to the diversion of steam away from the steam turbine, where up to 68% of the steam will be diverted for use in steam heating of the capture solvent to release the captured CO<sub>2</sub> for compression and transport. The plant's new net power will decrease from 1,104.46 MW down to 738.67 MW, due to the existing auxiliary power draw and the new power load used to power the added capture equipment. The resulting decrease in power output will lose the plant between \$206,793,038.55 and \$333,933,814.97 per year in power revenue profits, but can be made up by 45Q Tax Credit benefits totaling between \$223,103,199.69 and \$283,509,777.44 per year. The revenue short fall will exceed the amount of tax credits when the plant operates at a capacity factor greater than 84.9%, however this does not account for the price of the capture plant. When including the \$285 million price tag for all of the capture equipment, the highest average capacity factor achievable that will break even in cost will be 75.4539%. This presents a slight decrease in running the plant less than normal, given the current capacity factor for Blue Water is 78.6933%. Therefore, operating the plant at 78.6933% capacity will not cover the cost of adding the capture equipment, so a permanent 3.24% derate is required.

It is recommended that if DTE pursues carbon capture for Blue Water, that additional heat recovery strategies be put in place to further minimize the amount of steam diversion there is for solvent heating and regeneration. DTE will also need to identify a sequestration repository for their captured carbon, and how to transport it there. Transportation methods and cost were not included in the scope of this report. Currently, 45Q tax credits are the best means of making a profit off of adding carbon capture to an existing power plant, but it is important to note that the tax credits can only be earned for 12 years of the capture plant's operation. Long term strategy and planning will be required to make a carbon capture plant viable for the entire remaining life of Blue Water till 2052. It is important to decarbonize the power sector as rapidly as possible to lessen the impact of climate change, and until sufficient renewable generation and battery storage can penetrate the baseload power market, carbon capture is the fastest way towards cutting emissions. DTE should seriously consider adding carbon capture to their remaining long-term fossil fuel power plants including Blue Water, otherwise DTE will not be able to achieve their decarbonization and sustainability goals.

## 1.0 Introduction and Current State

DTE Energy (DTE) faces multiple challenges when addressing their current progress towards their 2050 net-zero carbon goals, including their existing fossil fueled power plants. Both remaining coal plants will be finally retired in 2028 (Belle River) and 2032 (Monroe) respectively, but no stated plans exist for reducing carbon emissions from DTE's natural gas based generation fleet (*Wilmot*). DTE's newest gas plant, the 1,150 MW-rated Natural Gas Combined Cycle (NGCC) Blue Water Energy Center (BWEC) is expected to last at least until 2052 (*EIA-860*), beyond DTE's net-zero deadline of 2050. DTE has publicly touted its pride in natural gas services and has no current plans for lessening its dependence on natural gas. To meet DTE's net zero needs and provide for the continuance of natural gas fueled service, this report will analyze the addition of carbon capture equipment to the plant and propose a properly sized system to be installed at BWEC. This system will be designed for 95% carbon capture at 95% purity, a significant improvement over the lack of carbon emissions control that BWEC has currently.

### 1.1 Overview of the Project

Blue Water Energy Center is the newest and largest natural gas plant in DTE's generation fleet. It sits on 40 acres of land east-adjacent to the older Belle River coal plant – which will be switching to natural gas use in 2026 – just north of the four Dean Peaker simple cycle natural gas units. Among small emissions of methane and NO<sub>x</sub>, BWEC releases an average of ~3,000,000 US tons of CO<sub>2</sub> into the atmosphere each year (*"2022 and 2023 Annual Power Plant Emissions."*). While cleaner burning than its nearby coal fueled neighbor to the west, BWEC contributes to the prominent carbon emissions and higher temperatures of East China Township, affecting the surrounding communities of Marine City, St. Clair, and the greater area of Port Huron. Adding carbon capture to BWEC would benefit the community and the State of Michigan by lessening the area's carbon footprint and contributing to a slower pace of climate change. Such equipment would also be a first for the State and could provide a template for future capture technology across DTE's fleet, allowing DTE to maintain its natural gas business while decreasing CO<sub>2</sub> as the largest contributor to global warming. Michigan is already considered a climate haven among environmentalists (*McWhirter*), so it is imperative to preserve Michigan as a State that maintains our environment in the face of climate change. Introducing carbon capture to BWEC would be of environmental and economic benefit to the public and DTE as well as socially responsible to the local community, if it can be implemented at the right price.

### 1.2 Review of Current State Design / Literature Review

Prior to this semester, I and fellow Master's Student Haylee Miller performed a techno-economic overview of the potential for carbon capture at Blue Water Energy Center. We determined that a slim business case existed based on our rough estimates for the cost of carbon capture implementation, highlighting that the maximum carbon tax credit would be necessary to balance out the large cost of adding the capture equipment. Specifically, we were able to roughly estimate the Levelized Cost of Electricity (LCOE) for BWEC both pre- and post- carbon capture retrofit and determined that a maximum \$85/ton CO<sub>2</sub> tax credit would cover the \$28.39/MWh rise in LCOE to \$73.66/MWh pre-tax benefit, reducing the overall plant LCOE from \$45.27/MWh pre-retrofit to

\$44.01/MWh post-retrofit. Examining the sizing analysis for the capture equipment was not in the scope of our techno-economic analysis at the time, which inspired this current project to refine the energy demand estimate as well as the cost of implementing carbon capture at BWEC.

Another Michigan Master's alumnus who has done such sizing analysis before is Caleb Smydra, performing a generic sizing analysis for a typical natural gas combined cycle plant for his ISD ESE capstone (*Smydra*). His hypothetical example plant was about half the size of BWEC (615 MW<sub>e</sub>) and operated less efficiently at a higher heat rate (7,500 Btu/kWh vs. 6,492 Btu/kWh reported for BWEC) (*Morren*). He designed only for 90% capture, resulting in a new total yearly emission of 270,000 tons of CO<sub>2</sub>. The result was a \$234 - \$453 million project situated on 5 to 10 acres of additional land added to the existing cost and site of the power plant (*Smydra*). Caleb and I both used numerous heat and energy balance equations from Heat Transfer and Chemical Engineering textbooks for the formulas used for each piece of equipment, much of which was sourced from Jennifer Wilcox's *Carbon Capture* textbook (*Wilcox*). Since Caleb's theoretical plant emits 2,700,000 tons annually with no carbon capture, at a capacity factor of 1, that means his plant expels 616,438.36 lb<sub>m</sub>/hr of CO<sub>2</sub> on average, so his system will be smaller than the one for BWEC, given that BWEC will be emitting a maximum average of 882,994.07 lb<sub>m</sub>/hr of CO<sub>2</sub>. I can expect that my sizing and estimates should be ~ 43% larger or more energy intensive than his theoretical plant. This is a good baseline with which to start my analysis.

### 1.3 Stakeholders

When constructing such a complex and wide-ranging solution – as is the case with implementing carbon capture – it is important to have a wide range of input from key stakeholders. The primary stakeholders are those directly involved and affected by the project, which in this case is mainly DTE and Blue Water plant personnel. Both the company, its financial stability, and those who work at Blue Water will be the most affected aspects of adding carbon capture to the plant – how the change in steam demand will affect the plant's operations, how the capture system will work and how it should be maintained, how much the project will cost, and how much payback DTE can expect in lowered taxes due to its sequestration efforts. This last part in particular has the ability to impact DTE's decision about its electricity rates, and how it will recover any losses in its bottom line as a result of adding this expensive equipment. While DTE is regulated by the Michigan Public Service Commission, DTE has certain leeway in determining its rates which affects consumers, particularly for residential electricity. The added need for fuel to compensate for less power production also affects DTE's purchase agreements with natural gas providers. Lastly, the general public – in particular the nearby communities surround East China Township – are also stakeholders in the success of the project, which will benefit from mitigating carbon emissions at Blue Water.

### 1.4 Scope

This project will only encompass a retrofit sizing of the carbon capture equipment necessary to sequester 95% of the total maximum carbon dioxide flow at 95% purity. A brief technical analysis will surround this sizing and engineering specification, stating the pre- and post-retrofit economic and operational profile which will include analysis of the plant's LCOE, Capacity Factor, and decrease in

maximum power output. Location siting will only be for the capture equipment, and any analysis and determination for sequestration & storage options will not be included in this project's scope.

### *1.5 Project Deliverables*

The three major goals of this report are to:

- 1) Deliver an industry quality report on implementing carbon capture at Blue Water Energy Center
- 2) To have an accurate measure of the following technical requirements:
  - a. A baseline, pre-retrofit engineering and economic operating profile of BWEC
  - b. An accurate estimate of carbon emissions at various operating profiles of BWEC
  - c. An accurate estimate of the auxiliary steam and electricity load required for capture
  - d. An accurate equipment sizing to capture 95% of BWEC's emissions at 95% purity
  - e. An accurate estimate of post-retrofit engineering and economic profile of BWEC
- 3) To provide industry quality analysis of the consequences of carbon capture implementation

In greater detail, my goal is to first determine the current state of BWEC: how it is currently operating with regard to maximum and average power output, capacity factor, its dispatch price (LCOE), and most importantly its carbon dioxide output. Next, the project will draw up a design and technical specifications for each major piece of carbon capture equipment, including – but not limited to – the Absorber (capture) Column, Regeneration (separation) Column, exhaust gas blower, capture solvent circulating pump(s), and CO<sub>2</sub> Compressor. Once the capture system's design is finalized, I can finish my analysis by determining the effect that adding the capture equipment will have on the overall power plant, including the new decreased maximum power output, the adjusted capacity factor to make up the lost power, and the new dispatch price (LCOE) of the plant, as well as the total reduction in carbon dioxide emissions.

There are seven key metrics that I have defined for this project that, when determined, will allow me to achieve the goals I listed above. These are the following:

1. The amount of carbon dioxide captured from a system designed for 95% capture at 95% purity
2. The decrease in steam sent to the steam turbine and subsequent loss in turbine power output
3. The sizing of each major component of the capture equipment
4. The amount of tax credits saved due to avoiding 95% of the total yearly carbon emissions
5. The new capital cost, O&M cost, and fuel cost in LCOE due to adding the capture system
6. The lowered LCOE of Blue Water once the carbon tax credits are applied to DTE's bottom line
7. The total decrease in maximum power output for BWEC post-retrofit, and the new operating capacity factor that will allow for optimal power generation and break even economic cost

These metrics are the critical pieces of data and information that I will need to use as comparison for the impact that implementing carbon capture will have, positive and negative, on the plant's operation.

## 1.6 Project Timing

The overarching schedule of the project and each deliverable are summarized below:

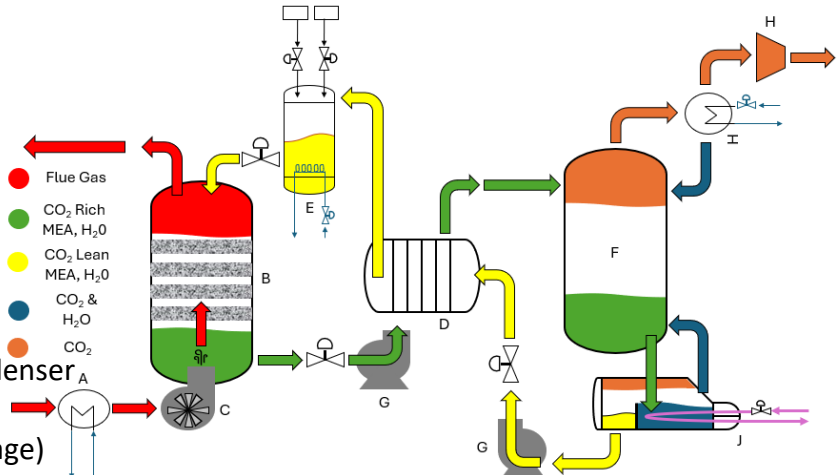
End of September (September 30<sup>th</sup>):

- Obtaining a project sponsor
- Data and information gathering
- Metric 1: Total Carbon Load and Amount Captured at 95% capture rate
  - Maximum expected CO<sub>2</sub> flow per HRSG: 441,497.03 lb<sub>m</sub> / hr
  - Total Maximum expected CO<sub>2</sub> flow of BWEC: 882,994.07 lb<sub>m</sub> / hr
  - Amount of CO<sub>2</sub> captured per hour at 95% capture: 838,844.37 lb<sub>m</sub> / hr
- Metric 2: Amount of Power Loss due to partial Steam Turbine steam diversion
  - Maximum derate of Steam Turbine – 301.23 MW
- Metric 4: Amount of 45Q Tax Credit Revenue saved (\$/year) through Carbon Capture
  - At \$12/ton CO<sub>2</sub>: \$31,496,919.83 / year, at \$85/ton CO<sub>2</sub>: \$223,103,182.14 / year

End of October (October 31<sup>st</sup>):

- Metric 3: Complete Sizing of Major Equipment for Carbon Capture system, including –

- A: Flue Gas Pre-Cooler
- B: Absorption Column
- C: Flue Gas Blower pump
- D: Amine Heat Exchanger
- E: Lean Amine Cooler
- F: Regeneration Column
- G: Solvent Pump(s)
- H: CO<sub>2</sub> Compressor
- I: CO<sub>2</sub> Pre-Compressor Condenser
- J: Rich Amine Reboiler
- Overall Site Footprint (acreage)



End of November (November 30<sup>th</sup>):

- All results, including system sizing, LCOE analysis, BWEC derate, and new capacity factor
- Metric 5: Estimates of BWEC's new capital cost, O&M cost, fuel cost, and LCOE
- Metric 6: BWEC's estimated LCOE after 45Q Tax Credits lowers plant operating costs
- Metric 7: BWEC's new derated max power output and Capacity Factor due to power make-up

December 9<sup>th</sup>: Project Report Submission



Table 2: Flow Stream Descriptions and Purpose

| Flow Stream: | Constituent Material:                       | Flow Stream Purpose:                                                                                                                                              |
|--------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a           | Flue gas                                    | Incoming gas turbine combustion flue gas redirected from existing HRSG stack(s) to capture system inlet                                                           |
| 1b           | Flue gas                                    | Flue gas after Pre-Cooler ("A"), cooled to match the temperature in the Absorption Column ("B") at 104°F (40°C)                                                   |
| 1c           | Flue gas                                    | Flue gas after Blower ("C"), is at increased pressure to overcome losses across the capture media in the Absorption Column ("B")                                  |
| 2            | CO <sub>2</sub> -Lean MEA, H <sub>2</sub> O | CO <sub>2</sub> -Lean MEA-H <sub>2</sub> O solvent mix leaving Lean Amine Storage Tank ("E") for Absorption Column ("B"), sprayed through capture media           |
| 3            | CO <sub>2</sub> -Rich MEA, H <sub>2</sub> O | CO <sub>2</sub> -Rich MEA-H <sub>2</sub> O solvent mix leaving Absorption Column ("B") for Amine Heat Exchanger ("D"), pressurized by Solvent Pump ("G")          |
| 4            | Flue gas                                    | Flue gas with low carbon constituency exiting the Absorption Column ("B"), bound for venting to atmosphere via HRSG stack(s)                                      |
| 5            | CO <sub>2</sub> -Rich MEA, H <sub>2</sub> O | CO <sub>2</sub> -Rich solvent mix leaving pre-heating Amine Heat Exchanger ("D"), ready for 248°F (120°C) conditions in Regeneration Column                       |
| 6            | CO <sub>2</sub> , H <sub>2</sub> O          | Condensed water – and some CO <sub>2</sub> – leaving from Condenser ("I"), used to ensure only pure CO <sub>2</sub> enters the Compressor ("H")                   |
| 7            | CO <sub>2</sub> -Rich MEA, H <sub>2</sub> O | CO <sub>2</sub> -Rich amine solvent sent from Regeneration Column ("F") to Reboiler ("J") for heated separation between MEA and CO <sub>2</sub>                   |
| 8            | CO <sub>2</sub> , H <sub>2</sub> O          | Regenerated CO <sub>2</sub> and H <sub>2</sub> O vapor returning to Regeneration Column ("F"), heated by Reboiler ("J") leaving Lean solvent mix behind           |
| 9            | CO <sub>2</sub> -Lean MEA, H <sub>2</sub> O | CO <sub>2</sub> -Lean MEA-H <sub>2</sub> O solvent mix leaving Regeneration Column ("F") for Amine Heat Exchanger ("D"), pressurized by Solvent Pump ("G")        |
| 10           | CO <sub>2</sub> -Lean MEA, H <sub>2</sub> O | CO <sub>2</sub> -Lean MEA-H <sub>2</sub> O solvent mix leaving from Amine Heat Exchanger ("D") to Lean Amine Storage Tank and Lean Amine Cooler ("E")             |
| 11           | CO <sub>2</sub>                             | CO <sub>2</sub> – with some H <sub>2</sub> O – exiting Regeneration Column ("F") as vapor, entering Condenser ("I") to remove moisture and purify CO <sub>2</sub> |
| 12           | CO <sub>2</sub>                             | Purified gaseous CO <sub>2</sub> leaving the Condenser ("I") and arriving at the Compressor ("H")                                                                 |
| 13           | CO <sub>2</sub>                             | Compressed CO <sub>2</sub> leaving the Compressor ("H"), ready for pressurized transport off-site                                                                 |
| 14           | H <sub>2</sub> O                            | Make-Up H <sub>2</sub> O needed to replenish any water that vaporizes and exits with flue gas in stream 4                                                         |
| 15           | MEA                                         | Make-Up MEA needed to replenish any solvent that vaporizes and exits with flue gas in stream 4                                                                    |

## 2.1 Flue Gas and CO<sub>2</sub> flow rates

Using emissions testing data publicly available on Blue Water Energy Center's (BWEC) Heat Recovery Steam Generators (HRSGs), I found that the maximum volume flow rate of flue gas exhausted per HRSG is reported to be between 1,335,998 ft<sup>3</sup>/min and 1,379,149 ft<sup>3</sup>/min (*Montrose*). From this, I estimated the maximum flow rate of CO<sub>2</sub> by taking the largest of the two values above, and using Eq. 1 to determine the hourly mass flow rate of CO<sub>2</sub> in lb<sub>m</sub>/hr:

$$\text{Eq. 1: } \dot{m}_{\text{mass flow rate}} \left( \frac{\text{lb}_m}{\text{hr}} \right) = \dot{V}_{\text{Volume flow rate}} \left( \frac{\text{ft}^3}{\text{min}} \right) * \text{vol \% } (\% \text{ CO}_2) * \text{density} \left( \frac{\text{lb}_m}{\text{ft}^3} \right) * 60 \frac{\text{min}}{\text{hr}}$$

For the maximum flow of flue gas at 1,379,149 ft<sup>3</sup>/min, the maximum constituent % by volume for CO<sub>2</sub> in the flue gas is 5.58% (*Montrose*), with a density at 168.9°F of ~ 0.0956 lb<sub>m</sub>/ft<sup>3</sup> (*Engineeringtoolbox*), the resulting mass flow rate is 441,497.0347 lb<sub>m</sub>/hr per HRSG. Doubling this value reveals the maximum expected hourly mass flow of CO<sub>2</sub> from the plant at 882,994.0695 lb<sub>m</sub>/hr, and taking 95% of that results in our design point for the capture system; capturing 838,844.366 lb<sub>m</sub>/hr of CO<sub>2</sub>, otherwise derived as 2,893,280.81 US tons/year. Solving for how many tons of CO<sub>2</sub> per year are captured utilizes the capacity factor (CF) for the plant and some unit conversions, as shown in Eq. 2:

$$\text{Eq. 2: } \dot{m}_{\text{yearly mass flow rate}} \left( \frac{\text{tons}}{\text{yr}} \right) = \dot{m}_{\text{hourly mass flow rate}} \left( \frac{\text{lb}_m}{\text{hr}} \right) * CF * 8,766 \frac{\text{hrs}}{\text{yr}} * \frac{1 \text{ ton}}{2,000 \text{ lb}_m}$$

The long-term capacity factor of BWEC is between 78% and 80%, higher than most NGCC plants (*EIA 2021*). This also means that BWEC consumes more fuel than the average NGCC plant, as it uses more fuel for longer periods of the year as it runs more often. Emissions data stated that the peak flow of natural gas, when operating at 100% gas turbine (CGT) and duct burner (DB) output, approached 45 lb<sub>m</sub>/s, or about 162,000 lb<sub>m</sub>/hr (*Montrose*). The molar flow rate of fuel – assumed to be pure methane – can be linked to the amount of CO<sub>2</sub> generated by pure combustion using Eq. 3:

$$\text{Eq. 3: } \dot{m}_{\text{CO}_2} \left( \frac{\text{lb}_m}{\text{s}} \right) = \dot{m}_{\text{CH}_4} \left( \frac{\text{lb}_m}{\text{s}} \right) * \frac{MW_{\text{CO}_2} \left( \frac{\text{lb}_m}{\text{mol}} \right)}{MW_{\text{CH}_4} \left( \frac{\text{lb}_m}{\text{mol}} \right)} * \frac{1 \text{ mol CO}_2}{1 \text{ mol CH}_4}$$

This relationship is also another way to determine the CO<sub>2</sub> emission rate for the plant. The capture system must be sized for the maximum emission rate possible, while the operating profile of the plant on a yearly basis must incorporate the capacity factor of the plant, as the plant never runs at 100% capacity for 100% of the year.

Lastly, taking all the data for the four major emitted gas constituents (each labeled *i*, representing CO<sub>2</sub>, O<sub>2</sub>, H<sub>2</sub>O, and N<sub>2</sub>) in the flue exhaust, it is possible to construct volume, mass, and molar flow rates for each and sum them to be representative of the total molar flue flow using Eq. 4:

$$\text{Eq. 4: } \dot{n}_{\text{flue gas}} \left( \frac{\text{mol}}{\text{s}} \right) = \frac{\dot{m}_i \left( \frac{\text{kg}}{\text{s}} \right)}{MW_i \left( \frac{\text{kg}}{\text{kmol}} \right) * x_i} * \left( \frac{1000 \text{ mol}}{1 \text{ kmol}} \right)$$

## 2.2 Flue Gas Pre-Cooler

Once the flow parameters of the flue gas is known, the cooling water demand for the Flue Gas Pre-Cooler can be calculated using Eq. 5 and Eq. 6 below (*Smydra*), which state the individual cooling load per each gas constituent (Eq. 5) and the total cooling load as the sum of each gas's required cooling load (Eq. 6):

$$\text{Eq. 5: Individual Cooling Load } Q_i \text{ (kW)} = \dot{m}_i \left( \frac{\text{kg}}{\text{s}} \right) * C_{p,i} \left( \frac{\text{kJ}}{\text{kg} * \text{K}} \right) * \Delta T (\text{K})$$

$$\text{Eq. 6: Total Cooling Load } Q_{total} \text{ (kW)} = \sum Q_i \text{ (kW)}$$

To calculate each constituent's individual cooling load, I solved for the mass flow rate of each gas within the flue stream ( $\dot{m}_i$ ) using Eq. 1 for H<sub>2</sub>O, O<sub>2</sub>, and N<sub>2</sub> in addition to CO<sub>2</sub>. This was followed by pulling reference numbers for the heat capacity of each gas ( $C_{p,i}$ ) (*Engineeringtoolbox*) and estimating the change in temperature ( $\Delta T$ ) from the maximum flue gas temperature at 184.9°F, down to 104°F (40°C) which matches up with the internal operating temperature of the Absorption Column. To solve for the amount of required cooling water, I used the total cooling load from Eq. 6 and applied Eq. 5 in reverse to solve for the required mass flow rate of cooling water needed for cooling down the flue gas. What resulted was a mass flow rate of water in kg/s which could be converted to gallons per minute (gpm).

## 2.3 Absorption Column

Once the flue gas has been cooled to match the Absorption Column's temperature, the flue gas is pressurized by a Blower and enters the Absorber. Before the Blower can be sized, the pressure drop across the Absorption Column must be determined. First, the Absorption Column must be fully sized. Using Eq. 4 as a basis, I calculated the average molar mass of the flue gas using Eq. 7:

$$\text{Eq. 7: } MW_{flue \text{ gas}} \left( \frac{\text{g}}{\text{mol}} \right) = \sum_i \frac{\dot{n}_i \left( \frac{\text{mol}}{\text{s}} \right)}{\dot{n}_{flue \text{ gas}} \left( \frac{\text{mol}}{\text{s}} \right)} * MW_i \left( \frac{\text{g}}{\text{mol}} \right)$$

Then, I used the Ideal Gas Law to determine the density (mass/volume) of the flue gas, using a gas constant of  $8.2057 \times 10^{-5} \text{ m}^3\text{-atm/mol-K}$ , a temperature of 313.15K (40°C), and the molar mass of the flue gas, computed to be 27.68 g/mol. Eq. 8 shows the modified Ideal Gas Law as:

$$\text{Eq. 8: } \rho_{flue \text{ gas}} = \frac{\dot{m}_{flue \text{ gas}} \left( \frac{\text{kg}}{\text{s}} \right)}{V(\text{m}^3)} = \frac{M_{flue \text{ gas}} \left( \frac{\text{g}}{\text{mol}} \right) * \frac{1 \text{ kg}}{1000 \text{ g}} * 1 \text{ atm}}{R \left( \frac{\text{m}^3 * \text{atm}}{\text{K} * \text{mol}} \right) * T(\text{K})}$$

Next, the abscissa value  $F_{LV}$  needed to be found in order to find the corresponding  $K_4$  value. Literature review highlighted that the corresponding density of the MEA solvent was 1,041.3 kg/m<sup>3</sup>, and an average liquid to gas ratio of 2.7 would be a good setpoint (*Smydra*). Applying this with Eq. 9 (*Wilcox*) is as follows:

$$Eq. 9: F_{LV} = \frac{L \left( \frac{mol}{s} \right)}{G \left( \frac{mol}{s} \right)} * \sqrt{\frac{\rho_G \left( \frac{kg}{m^3} \right)}{\rho_L \left( \frac{kg}{m^3} \right)}}$$

Taking this new  $F_{LV}$  value of  $\sim 0.0868$  with the assistance of the generalized pressure drop correlation (Appendix A), the corresponding  $K_4$  value of 0.8 was determined. Using this new value, it is possible to apply Eq. 10 (*Wilcox*) to solve for  $V_w^*$ , the gas mass flow rate per unit column cross-sectional area, which determines the flow rate per unit area that is possible for the constraints of the column. A few new quantities appear, including the Packing Factor ( $F_P = 21$ ) of the capture media and the viscosity of the MEA solvent, which is reported to be 0.0013 Pa-s (*Smydra*). Other values, including the densities of the flue gas and MEA solvent, reappear from the previous equation. Thus:

$$Eq. 10: K_4 = \frac{13.1 * V_w^{*2} \left( \frac{kg}{m^2 * s} \right) * F_P * \left( \frac{\mu_L (Pa * s)}{\rho_L \left( \frac{kg}{m^3} \right)} \right)^{.1}}{\rho_G \left( \frac{kg}{m^3} \right) * \left( \rho_L \left( \frac{kg}{m^3} \right) - \rho_G \left( \frac{kg}{m^3} \right) \right)}$$

By solving for  $V_w^*$ , it is possible to obtain the cross-sectional area of the Absorption Column via Eq. 11:

$$Eq. 11: Area_{absorption\ column} (ft^2) = \frac{\dot{m}_{flue\ gas} \left( \frac{kg}{s} \right)}{V_w^* \left( \frac{kg}{m^2 * s} \right)}$$

$$Eq. 12: Diameter_{absorption\ column} (ft) = 2 * \sqrt{\frac{Area (ft^2)}{\pi}}$$

Lastly, applying the formula (Eq. 12) for the area of a circle will deliver the diameter for the given mass flow. However, if the diameter exceeds that of buildable standards ( $> 15$  m), then two or more Absorption Columns handling half or less of the total flue flow will be required (*Wilcox*).

As for the height of the Absorption Column, there are multiple chemical engineering terms to calculate that allow for the computation of the total height of the column. These include the molar ratio of  $CO_2$  to the total molar flow at the inlet and outlet of the Absorber (Eq. 13), the number of mass-transfer units (NTU) known as  $N_{OG}$  (Eq. 14), the height of each mass transfer unit  $H_{OG}$  (Eq. 15), and the product of  $N_{OG}$  and  $H_{OG}$  to make up the total height of the column  $Z_T$  (Eq. 16). Here are the equations in detail:

$$Eq. 13: y = n_{CO_2} / n_{total}$$

$$Eq. 14: N_{OG} = \frac{\ln \left( \left( 1 - m \left( \frac{V}{L} \right)_m \right) * \frac{y_{in}}{y_{out}} + m \left( \frac{V}{L} \right)_m \right)}{\left( 1 - m \left( \frac{V}{L} \right)_m \right) * \eta_{stage}}$$

$$Eq. 15: H_{OG} = H_G + m \left( \frac{L}{V} \right)_m * H_L$$

$$Eq. 16: Z_T = 1.15 * N_{OG} * H_{OG}$$

Using the  $N_{OG}$  equation is similar to the McCabe-Theile manual stage analysis process but involves less graphical analysis and the inherent inaccuracy involved with it. Of note, Eq. 14 includes the 20% stage efficiency ( $\eta_{stage}$ ) which represent the inefficiency of capturing  $CO_2$  within each stage of

MEA and capture media (*Smydra*). The literature review from Wilcox recommended a value of  $m(V/L)_m$  between 0.7 and 0.8 when computing the  $N_{OG}$  value, and the inverse of 0.7-0.8 for the  $m(L/V)_m$  value which is used to compute the height of each stage ( $H_{OG}$ ). Applying all four equations together results in a total stage height of  $Z_T$  (Wilcox), which includes the extra 15% height for the header and footer of the column.

## 2.4 Flue Gas Blower

The key metric with the Flue Gas Blower is that it is able to pressurize the flue gas such that it can overcome any pressure losses as the flue gas flows across the capture media of the Absorption Column. A Blower was chosen because of the need to overcome a pressure drop greater than the capacity of a fan (15 kPa limit). To design the Blower successfully, first it is necessary to determine the pressure drop across the height of the column. This is found by taking the pressure at the flue gas exit and adding the product of the Absorption Column height and the pressure drop per length of packing material, shown in Eq. 17:

$$\text{Eq. 17: } \text{Pressure}_{\text{inlet}}(\text{atm}) = \text{Pressure}_{\text{outlet}}(\text{atm}) + \text{Height}_{\text{Absorption Column}}(\text{ft}) * \Delta \text{Pressure} \left( \frac{\text{atm}}{\text{ft}} \right)$$

The next step is to incorporate the inlet ( $p_2$ ) and outlet pressure ( $p_1$ ) with the temperature of the Absorption Column ( $T$ ), the efficiency of the Blower ( $\varepsilon = 0.65\text{-}0.85$ , 0.65 was chosen to compute the maximum power required), the flow rate of the flue gas ( $\dot{m}$ ) and the average molar weight ( $MW$ ), and the specific heat ratio ( $k$ ) – assumed to be 1.4 – using Eq. 18 (Wilcox) to obtain the power demand of the Blower:

$$\text{Eq. 18: } \text{Power (kW)} = \left( \frac{\dot{m}_{\text{flue gas}} \left( \frac{\text{mol}}{\text{s}} \right) * R \left( \frac{\text{KJ}}{\text{mol} * \text{K}} \right) * T(\text{K})}{MW_{\text{flue gas}} * \varepsilon_{\text{blower}}} \right) * \left( \frac{k}{k-1} \right) * \left( \left( \frac{p_2(\text{atm})}{p_1(\text{atm})} \right)^{\frac{k-1}{k}} - 1 \right)$$

Finally, the last step to find the volume demand of the Blower is found using the Ideal Gas Law ( $P * V = \frac{\dot{m}}{MW} RT$ ) using the following modification in Eq. 19:

$$\text{Eq. 19: } V(\text{m}^3) = \frac{\dot{m}_{\text{flue gas}} \left( \frac{\text{kg}}{\text{s}} \right)}{M_{\text{flue gas}} \left( \frac{\text{g}}{\text{mol}} \right) * \frac{1 \text{ kg}}{1000 \text{ g}}} * \frac{R \left( \frac{\text{m}^3 * \text{atm}}{\text{K} * \text{mol}} \right) * T(\text{K})}{P(\text{atm})}$$

## 2.5 Amine Heat Exchanger

The key attributes for the design of the Amine Heat Exchanger are the surface heating area of the exchanger and the thermal power demand. The temperatures that are needed to be achieved across the heat exchanger determine the thermal demand and also determine how much surface area is needed to achieve sufficient heat transfer. The  $\text{CO}_2$  Rich inlet temperature (stream 3) is known from previously at 40°C but increased to 60.68°C due to the exothermic reaction of the MEA binding to the  $\text{CO}_2$ , releasing 85 kJ/mole of  $\text{CO}_2$  captured. The outlet from the Reboiler to the inlet of the lean solvent (stream 9) was assumed to be 120°C. The approach temperature for the exiting Lean  $\text{CO}_2$  (stream 10) is limited to 11°C, adding that to the existing incoming temperature for stream 3 results in a 71.68°C

approach. Using Eq. 5, it is possible to solve for the exiting temperature of the Rich CO<sub>2</sub> (stream 5) by solving for the total thermal power demand and reverse solving for the temperature change of the Rich CO<sub>2</sub> stream. Finally, using both Eq. 20 and 21, I solved for the heat transfer surface area by converting the kW of thermal load into Btu/hr and solving for the surface area (A) using Eq. 21:

$$\text{Eq. 20: } \Delta T_{LM} (^{\circ}\text{F}) = \frac{(T_{H1} (^{\circ}\text{F}) - T_{C2} (^{\circ}\text{F})) - (T_{H2} (^{\circ}\text{F}) - T_{C1} (^{\circ}\text{F}))}{\ln \left( \frac{T_{H1} (^{\circ}\text{F}) - T_{C2} (^{\circ}\text{F})}{T_{H2} (^{\circ}\text{F}) - T_{C1} (^{\circ}\text{F})} \right)}$$

$$\text{Eq. 21: } Q \left( \frac{\text{BTU}}{\text{hr}} \right) = U \left( \frac{\text{BTU}}{\text{hr} \cdot \text{ft}^2 \cdot ^{\circ}\text{F}} \right) * A (\text{ft}^2) * \Delta T_{LM} (^{\circ}\text{F})$$

Through literature review, a suitable heat transfer coefficient ( $U$ ) of 200 Btu/hr-ft<sup>2</sup>-°F could be assumed for a water and water-organic fluid heat exchange process (*Smydra*).

## 2.6 Lean Amine Cooler

Embedded in the tank storing the CO<sub>2</sub>-Lean MEA-water solution that gets sprayed into the Absorption Column, the Lean Amine Cooler maintains the temperature in the tank such that the outlet flow (stream 2) is at the same temperature as the Absorption Column (40°C). Knowing the approach temperature of the incoming Lean MEA solution (stream 10) from above (71.68°C) and outgoing MEA (stream 2, 40°C), applying Eq. 5 solves for the total cooling demand. Setting Eq. 5 equal to the solved for cooling demand and reversing the equation to solve for the mass flow rate of cooling water is possible, given that the cooling water has a heat capacity of 4.18 kJ/kg-K (*Engineeringtoolbox*).

## 2.7 Regeneration Column and Reboiler

Sizing the Regeneration Column is similar to the process used for the Absorption Column. The setpoint temperature of the Regeneration Column is 120°C. Using the computed molecular weight of exiting CO<sub>2</sub>-H<sub>2</sub>O vapor (stream 11), the density of the flow was solved by using the same variation of the Ideal Gas Law used previously (Eq. 8). Following this, the L/G ratio can be found and applied to Eq. 9, solving for the  $F_{LV}$  value. Using the generalized pressure drop correlation (Appendix A) the  $K_4$  value is found and applied to Eq. 10 to find the  $V_w^*$ , which is then applied to solve for the area of the Regeneration Column using Eq. 11. Finally, the area is used to deduce the diameter of the column using Eq. 12, which also carries the same limiting size requirements (must be < 15m wide).

As for the height of the Regeneration Column, literature review recommended that the number of gas transfer units be between 15-20 stages, which is about 5x – or divided by 20% – greater than the typical computed number of staging that the McCabe-Thiele process predicts (*Smydra*). Eq. 13-16 are reused during this sizing process, which also includes the same 115% of  $Z_T$  to account for head flow at either end of the Regeneration Column. The last requirement of the height sizing is that the height must be at a suitable ratio to the diameter of the column, such that the height ratio ranges within 2-5 times the diameter.

Additionally, the Reboiler – which helps heat the Regeneration Column and add heat energy to separate the CO<sub>2</sub> from the MEA – has a significant steam demand for its heating purposes, which can be broken down into three components: the sensible heat load, the desorption load, and the

vaporization load. First, the sensible heat load is required to raise any liquid water in the Reboiler to saturated steam temperature of 120°C. Second, the desorption load is the heat requirement necessary to desorb the CO<sub>2</sub> from the MEA, which is slightly greater than the energy of absorbing the CO<sub>2</sub>. Third, the vaporization load is the heat demand necessary to vaporize any water and return it to the Regeneration Column along with the gaseous CO<sub>2</sub>, leaving behind the CO<sub>2</sub>-Lean amine solvent. This can be formalized into one total steam heating demand by Eq. 22:

$$\text{Eq. 22: } Q_{\text{reboiler}} = Q_{\text{sensible}} + Q_{\text{desorption}} + Q_{\text{vaporize}}$$

Solving for each quantity requires use of Eq. 5 again for  $Q_{\text{sensible}}$ , the heat of regeneration for  $Q_{\text{desorption}}$ , and the product of the mass flow rate of steam and the vaporization energy of water for  $Q_{\text{vaporize}}$ . After this, the Reboiler's steam demand is known and can be subtracted from the total steam flow sent to the steam turbine, which at maximum flow sends 2,082 thousand pounds per hour (kpph) of steam to the turbine to generate up to 443 MW of power.

## 2.8 Process Make-Up

Both MEA and H<sub>2</sub>O have the capacity to vaporize during multiple parts of the capture process and exit the system either in the flue gas leaving the Absorption Column, or as part of the capture stream of CO<sub>2</sub>. Having to add additional solvent into the holding tank is not optimal, but these losses will only occur in small quantities which minimizes their replacement cost. This analysis decided against estimating exactly how much of each molecule will leave the system, but literature research indicates that pilot carbon capture plants experience MEA losses anywhere from 0.5-3.1 kg MEA/metric ton of CO<sub>2</sub> captured, and water losses estimated around 156 gpm (*Smydra*).

## 2.9 Solvent Pump(s)

In order to overcome elevation differences and to keep the MEA solvent moving throughout the system, pumps are needed to pressurize the pipes and keep the rich and lean solvent flowing. To do so, the equation to use is Eq. 23 which incorporates volume flow rate ( $\dot{V}$ ), differential pressure ( $\Delta P$ ), and the intrinsic efficiency of the pump ( $\varepsilon_i$ ) as well as the efficiency of the driver motor ( $\varepsilon_d$ ):

$$\text{Eq. 23: } P_{\text{pump}}(kW_e) = \frac{\dot{V} \left( \frac{m^3}{s} \right) * \Delta P (Pa)}{\varepsilon_i * \varepsilon_d}$$

Wilcox states that the intrinsic pump efficiency can range from 40% to 85%, while the driver motor typically has a flat efficiency of 85% (*Wilcox*). To plan for the maximum power demand, I assumed the intrinsic pump efficiency was 40% and estimated a head pressure that needed to be overcome of 30 psig (recommended by *Smydra*).

## 2.10 CO<sub>2</sub> Pre-Compression Condenser

Before the CO<sub>2</sub> can be pressurized and released for transport, the CO<sub>2</sub>-H<sub>2</sub>O vapor mix must have its water removed from the CO<sub>2</sub> in a drying process that is industrially well-known for use in purifying

natural gas. This is where the Pre-Compression Condenser comes in. Typical systems use triethylene glycol as a desiccant to dry out water vapor from natural gas, but here instead a temperature approach is used to chill the water into a liquid, naturally removing the H<sub>2</sub>O before the gaseous CO<sub>2</sub> reaches the compressor. Sizing a condenser using Eq. 5, 20, and 21 determines both the thermal load of cooling water needed to chill the water vapor below its boiling point of 100°C, and the amount of heat exchanger surface area needed to ensure suitable heat transfer.

## 2.11 CO<sub>2</sub> Compression

Once the CO<sub>2</sub> has been purified – removing any remaining water vapor from the gas stream – it is ready for compression in preparation for transportation to its sequestration destination. Typical shipping and pipeline pressures call for 120 atmospheres of pressure prior to transport (*Smydra*). To reach such high pressure, successive compressive staging will be required. Due to this, a typical industrial compressor would need inter-stage cooling to combat the compressive heating of the gas as it moves through the stages. To determine the total heating experienced by the CO<sub>2</sub>, Eq. 24 was used after determining a set pressure ratio for each stage of the compressor:

$$\text{Eq. 24: } T_{out}(K) = T_{in}(K) * r_{stage}^{\frac{k-1}{k}}$$

As a result, the CO<sub>2</sub> would be chilled to counter the heating experience during compression. Eq. 25 is applied first to determine the number of stages using the total compressor pressure ratio of 120 atm over 2.1288 atm as follows (*Wilcox*):

$$\text{Eq. 25: } r_{stage} = \sqrt[\text{\# of stages}]{r_{total}}$$

According to Wilcox, the pressure ratio of each stage ( $r_{stage}$ ) should be less than 4 to avoid excessive heating of the compressed gas, which is why a 2.74 compression ratio was chosen. Each stage would need a different amount of power based on the temperature it was at and the compressibility factors across the stage. To determine the compressibility factor, both the reduced temperature and reduced pressure must be computed for each part of each stage, using Eq. 26 and Eq. 27 (*Smydra*):

$$\text{Eq. 26: } T_r = \frac{T(K)}{T_c(K)}$$

$$\text{Eq. 27: } P_r = \frac{P(atm)}{P_c(atm)}$$

This leads to Eq. 28 from Wilcox, which uses all of the combined information to compute the power requirement of each stage of the compressor (*Wilcox*):

$$\text{Eq. 28: } P(kW_e) = \left( \frac{n_{gas} \left( \frac{mol}{s} \right) * R \left( \frac{J}{mol * K} \right) * T_{in}(K)}{E} \right) * \left( \frac{k}{k-1} \right) * \left( r_{stage}^{\frac{k-1}{k}} - 1 \right) * \left( \frac{Z_1 + Z_2}{2 * Z_1} \right)$$

The standard universal gas constant of 8.314 J/mol-K was used alongside a compressor efficiency of 85% (lower end of range from Wilcox) and a specific heat ratio of 1.29 (*Smydra*).

## 2.12 Total Site Footprint

While much of the major tank and column equipment has a known cross-sectional area, the various pumps, pipes, heat exchangers, and cooling tower do not have a set footprint. Literature review indicated that 5 to 10 acres of land is needed for all the additional capture equipment. Suitable acreage was mapped using Google Earth around the surrounding area of the plant, being mindful of constraining features such as the westward 345 kV transmission lines. The resulting areas were chosen to minimize the length of pipe runs between each of the major equipment pieces and maintain space around the existing natural gas plant.

## 3.0 Results and Discussion

### 3.1 Absorption Column Results

The Absorption Column was sized at 100% maximum flue gas flow, representing 1,379,149 ft<sup>3</sup>/min of flue gas flow per HRSG stack with up to 5.58% of carbon dioxide by volume. At a density of 0.0956 lb<sub>m</sub>/ft<sup>3</sup> for the given minimum flue gas temperature of 163°F (winter exit conditions), this meant a carbon dioxide flow of 441,497.03 lb<sub>m</sub>/hour per HRSG stack, or a combined flow of 882,994.07 lb<sub>m</sub>/hour of CO<sub>2</sub>.

Given the constituent gases in the flue gas, their molecular weights and molar ratios, a molar mass of the flue gas was determined to be 28.42 g/mol, with a density of 1.2456 kg/m<sup>3</sup>. Eq. 9 yielded a  $F_{LV}$  value of 0.0934, which correlates to a  $K_4$  value of 0.8 using the  $F_{LV} - K_4$  chart in Appendix A, which was used to find the  $V_w^*$  value of 3.83 kg/m<sup>2</sup>-s using Eq. 10.  $V_w^*$  represents the flue gas mass flow rate per unit column cross-sectional area, tying together the rate of reaction and sizing of the area of the column, which was determined using Eq. 11 to be an initial value of 332.21 m<sup>2</sup> with diameter of 20.57 m – found using Eq. 12. This first design pass only used 1 total Absorption Column for the combined flow of both HRSG stacks and flue gas, which had a diameter that exceeded the buildable standard of 15 m across. Recalculating with the same mass flow of combined flue gas – 1,272.38 kg/s – but split between 2 Absorber Columns handling half the total flow resulted in each column area of 166.11 m<sup>2</sup> (1,787.96 ft<sup>2</sup>) and 14.54 m (47.71 ft) in diameter. This means that the most reasonable design of the Absorber Columns requires one column per HRSG exhaust stack and can only handle half of the maximum flue gas flow.

As for the height of the column, molar analysis of the mole fraction of CO<sub>2</sub> in the flue gas before and after capture indicated an in-out ratio ( $y_{in}/y_{out}$ ) of 19.1315 with Eq. 13. Using a  $m(V/L)_m$  value of 0.7 (and thus a  $m(L/V)_m$  value of 1.4286) from Wilcox, calculating the  $N_{OG}$  and  $H_{OG}$  using Eq. 14 and 15 resulted in 31.0407 stages (at 20% efficiency each) and a height per stage of 1 m (*Wilcox*). Adding 15% additional header room resulted in a column that is 35.7 m (117.12 ft) tall. This resulted in a height/diameter ratio of 2.45, which is within the recommended value of 2-5. The pressure drop across the column was 0.126178 atm from the outlet of the Blower to the exit of the flue gas back to the HRSG stacks – based on a pressure drop per stage of 42 mm H<sub>2</sub>O (pressure)/meter (*Smydra*).

Lastly, the best estimates for pricing of the Absorber Columns depends on the total volume of packed media within each column. Random packing can cost \$6 - \$100 per cubic foot, whereas structured packing can cost from \$45 - \$400 per cubic foot (*Smydra*). For a packing volume of 182,084.75 ft<sup>3</sup>, the maximum cost per column would be \$72,833,898.92, or \$145,667,797.84 for both required Absorber Columns. A summary of the results for the Absorber Columns is given below in Table 3:

Table 3: Absorber Column Results

| Number of Columns | 2                                  | each handle ½ of capacity             |
|-------------------|------------------------------------|---------------------------------------|
| Column Diameter   | 47.71   14.54                      | ft   m                                |
| Column Height     | 117.12   35.70                     | ft   m                                |
| Pressure Drop     | 0.126178   42                      | atm   mmH <sub>2</sub> O/m of packing |
| Packing Volume    | 182,084.75   5,156.07              | ft <sup>3</sup>   m <sup>3</sup>      |
| Maximum Cost      | \$72,833,898.92   \$145,667,797.84 | \$/Column   \$ for Both Columns       |

### 3.2 Regeneration Column Results

The Regeneration Column was sized at an incoming flow rate of 1,410,498.239 lb<sub>m</sub>/hr (177.72 kg/s) of CO<sub>2</sub> and an outgoing return flow rate of 571,675.603 lb<sub>m</sub>/hr (72.03 kg/s) of CO<sub>2</sub> leaving for the Lean Amine Storage Tank. This means that the true calculated flow of captured CO<sub>2</sub> – 105.69273793 kg/s – is equal to the capture target number of 838,844.37 lb<sub>m</sub>/hr of CO<sub>2</sub> and is separated from the MEA solution and sent to the condenser and compressor bound for transport. This stream is 95% CO<sub>2</sub> and 5% water vapor, with a molar mass of 42.7 g/mol and a density of 1.3236 kg/m<sup>3</sup> at 120°C.

Applying Eq. 9, given the above values for the exiting capture gas, yielded a  $F_{LV}$  value of 1.7318, which correlates to a  $K_4$  value of 0.2 using the  $F_{LV} - K_4$  chart in Appendix A. Eq. 10 calculated the  $V_w^*$  value of the column was 1.974 kg/m<sup>2</sup>-s and Eq. 11 determined the area of the column needed to be 54.7 m<sup>2</sup> (588.74 ft<sup>2</sup>) with diameter of 8.35 m (27.38 ft) – again, found using Eq. 12. This design met the < 15m diameter buildable constraint, which means that the Regeneration Column only requires one column total which can handle the maximum flow of CO<sub>2</sub>-Rich amine solution fed to it from both Absorber Columns. As for the height of the column, molar analysis of the mole fraction of CO<sub>2</sub> in the rich solvent stream and the 95% CO<sub>2</sub> capture stream indicated an in-out ratio ( $y_{out}/y_{in}$ ) of 28.8878, found using Eq. 13. Using a  $m(V/L)_m$  value of 0.7 (and thus a  $m(L/V)_m$  value of 1.4286) from Wilcox, calculating the  $N_{OG}$  and  $H_{OG}$  using Eq. 14 and 15 resulted in 37.2854 stages (at 20% efficiency each) and a height per stage of 0.85 m (*Wilcox, Smydra*). Adding 15% additional header room resulted in a column that is 36.45 m (119.57 ft) tall. This resulted in a height/diameter ratio of 4.37, which is within the recommended value of 2-5. The pressure drop across the column was 0.128828 atm from the outlet of the capture stream at the top of the column to the exit of the Reboiler at the bottom of the column where the flow is headed back to the Amine Heat Exchanger and Lean Amine Storage Tank – again, this pressure drop was based on a pressure drop per stage of 42 mm H<sub>2</sub>O (pressure)/meter (*Smydra*).

Next, the Reboiler steam heating demand was determined using each portion of the heat load. The sensible heat load was found using Eq. 5 to be 135,257.95 kW<sub>thermal</sub>, raising the incoming liquid temperature from 107.2354°C at the Amine Heat Exchanger outlet to 120°C at the Regeneration Column. The desorption heat load, separating the bound CO<sub>2</sub> and MEA, was assumed to be 100

kJ/mole of CO<sub>2</sub>, totaling 67,117.71 kW<sub>thermal</sub>. The vaporization head load was found to be 370,177.70 kW<sub>thermal</sub>, heating the saturated liquid water to saturated steam assuming a vaporization heat of 2,200.36 kJ/kg. Together, the total Reboiler heat demand is 572,553.36 kW<sub>thermal</sub>, requiring 178.38 kg/s of main plant steam to heat the 2,768.69 kg/s of liquid water and CO<sub>2</sub>-MEA falling to the bottom of the column. Though the plant's main steam operates at 600°C (1,112°F) and 180 bar (2,610.68 psi), it is more advantageous to divert the required steam amount and expand the pipe diameter to take advantage of lower pressure thermodynamic properties. Reducing the steam down to 2.685 atm pressure, the 178.38 kg/s of steam equates to 1,415.72 kpph, which at its max operating point is 68% of the total steam generation capacity of the plant. This leaves the steam turbine 666.28 kpph of steam left for generating power, producing 141.77 MW. The steam turbine therefore experiences a 301.23 MW derate, which is highly significant. This represents a loss of power over 4 times that of the auxiliary power demand for the entire capture system. This power derate due to diverted steam is not to be taken lightly, as it will have substantial operational and financial consequences for Blue Water and DTE. The Reboiler results are summarized in Table 5.

Lastly, the best estimates for pricing of the Regeneration Column again depends on the total volume of packed media within each column. Random packing can cost \$6 - \$100 per cubic foot, whereas structured packing can cost from \$45 - \$400 per cubic foot (*Smydra*). For a packing volume of 61,216.51 ft<sup>3</sup>, the maximum cost per column would be \$24,486,604.33. A summary of the results for the Regeneration Column is given below in Table 4:

Table 4: Regeneration Column Results

|                   |                      |                                       |
|-------------------|----------------------|---------------------------------------|
| Number of Columns | 1                    | handles full system capacity          |
| Column Diameter   | 27.38   8.35         | ft   m                                |
| Column Height     | 119.57   36.45       | ft   m                                |
| Pressure Drop     | 0.128828   42        | atm   mmH <sub>2</sub> O/m of packing |
| Packing Volume    | 61,216.51   1,733.46 | ft <sup>3</sup>   m <sup>3</sup>      |
| Maximum Cost      | \$24,486,604.33      | \$ for one single column              |

Table 5: Reboiler Steam Demand

|                               |                   |                                    |
|-------------------------------|-------------------|------------------------------------|
| Total Heating Demand          | 572,553.36        | kW thermal load                    |
| Total HRSG Steam Demand       | 178.38   1,415.72 | kg/s   kpph                        |
| Steam Turbine Maximum Derate  | 301.23   68%      | MW derate   % of total power       |
| Steam Turbine Available Power | 141.77   666.28   | MW available   kpph remaining      |
| Total Cost                    | \$1,847,737.41    | \$ for one Reboiler heat exchanger |

### 3.3 Flue Gas Blower Results

The initial results for the Flue Gas Blower called for a 16,574.42 kW<sub>e</sub> (22,226.63 hp) blower handling 1,031.5 m<sup>3</sup>/s (2,164,438.49 cfm) of flue gas. However, the largest industrial fans can only handle 1,500,000 cfm powered at 15,000 hp, and the flue gas from each HRSG stack will be passed individually to each Absorber Column. While crossover piping might be a nice redundant feature to

have, it makes more sense to have two Flue Gas Blower fans instead of a single one, in light of the above information. Therefore, the final results for the Flue Gas Blower were divided in two and are given below in Table 6. Additionally, the pricing of each blower is based on the cross-sectional area of the column that it will be servicing. The total cost for both blowers can also be found in Table 6:

Table 6: Flue Gas Blower Results

|                                              |                               |                                           |
|----------------------------------------------|-------------------------------|-------------------------------------------|
| $\Delta P_{blower}$ across Absorption Column | 0.1261783   1.8543            | atm   psi                                 |
| Total Power Load                             | 8,287.21   11,113.31          | kW / blower   hp / blower                 |
| Volumetric Flowrate                          | 1,082,219.24   510.75         | cfm / blower   m <sup>3</sup> /s / blower |
| Total Cost                                   | \$934,416.06   \$1,868,832.12 | \$ / blower   \$ for Both Blowers         |

### 3.4 Amine Heat Exchanger Results

The Rich/Lean Amine Heat Exchanger has a high thermal load required to preheat the CO<sub>2</sub>-Rich amine stream from 60.68°C to 107.24°C, approaching the 120°C temperature of the Regeneration Column. On the reverse side, the CO<sub>2</sub>-Lean stream must be cooled from 120°C to 71.68°C, approaching the 40°C nominal temperature of the Lean Amine Cooling Tank. The Lean Amine Cooler inside the tank takes care of the rest of the decrease in temperature to 40°C. The heat exchange load was found to be 453,422.78 kW<sub>thermal</sub> (1,547.0785 MMBtu/hr), requiring 92,561.81 ft<sup>2</sup> (8,599.28 m<sup>2</sup>) of heat exchanger surface area. One limitation of heat exchanger design is that some designs may be limited by the surface area they can support; for example, common shell and tube heat exchangers are typically limited to 12,000 ft<sup>2</sup>, which would mean the capture system would need at least 8 shell and tube heat exchangers to meet the heat load requirement. Alternatively, large plate and frame style or coil wound heat exchangers can achieve surface areas up to 375,000 ft<sup>2</sup>, which would more than fit the needs of the capture system. The cost for the Rich/Lean Amine Heat Exchanger was found to be \$6,730,931.68. All of these results are summarized below in Table 7:

Table 7: Amine Heat Exchanger Results

|                                      |                         |                                  |
|--------------------------------------|-------------------------|----------------------------------|
| Heat Load                            | 453,422.78   1,547.0785 | kW thermal   MMBtu/hr            |
| Required heat exchanger surface area | 92,561.81   8,599.28    | ft <sup>2</sup>   m <sup>2</sup> |
| Total Cost                           | \$6,730,931.68          | \$ for one Amine Heat Exchanger  |

### 3.5 Amine and Water Pump Results

The same basic process was used to analyze each pump, assuming a standard 30 psi pressure drop to overcome and a minimum intrinsic pump efficiency of 40% (pump efficiencies can range from 40% to 85%) and drive motor efficiency of 85%. Converting the mass flow rate of each stream into a volume flow rate allows for the use of Eq. 23 to compute the power requirement of the pump. Assuming a flat \$1,600/kW of power requirement for each pump, each pump's specific capital cost could be calculated. The power requirement and cost of each pump is given below in Table 8:

Table 8: Amine and Water Pump Results

| $\Delta P_{pump}$ estimate          | 30                        | psi                                                |
|-------------------------------------|---------------------------|----------------------------------------------------|
| Pump Efficiencies                   | 40%   85%                 | Intrinsic pump efficiency   Drive motor efficiency |
| Rich Amine Solvent Pump             | 1,541.63   \$2,466,604.52 | kW   \$ of pump                                    |
| Lean Amine Solvent Pump             | 1,586.26   \$2,538,011.30 | kW   \$ of pump                                    |
| Lean Amine Cooler Tank Pump         | 5,411.76   \$8,658,820.76 | kW   \$ of pump                                    |
| Flue Gas Pre-Cooler Pump            | 1,060.68   \$1,697,084.41 | kW   \$ of pump                                    |
| Compressor Inter-Stage Cooling Pump | 492.04   \$787,268.36     | kW   \$ of pump                                    |
| Regeneration Column Condenser Pump  | 187.76   \$300,412.66     | kW   \$ of pump                                    |

### 3.6 Compressor Results

Pipeline or other transport of CO<sub>2</sub> requires high pressures to compress ambient temperature CO<sub>2</sub> to liquid state for ease of flow and injection into sequestration endpoints. Specifically, literature recommended 120 atm (1,763.51 psia) pressure, requiring a 56.369 total compression ratio. This was achieved through 4 stages of compression at a stage compression ratio of 2.74, for a CO<sub>2</sub> mass flowrate of 105.69 kg/s (838,844.37 lb<sub>m</sub>/hour). Including times when the plant will run a partial power generation load, represented by the 78.6933% capacity factor for the plant, the plant will capture 2,624,743.32 metric tons per year (2,893,280.81 US tons per year) out of the 2,769,473.65 metric tons per year (3,052,821 US tons per year) emitted last year in 2023. This represents 94.774% of the total emissions captured, which reflects the goal of 95% capture at 95% purity. To remove the 5% water from the capture stream to achieve 100% purity of CO<sub>2</sub>, an additional desiccant system would need to be designed and installed prior to the compressor, which was out of scope for this project. The compressor will require 35,292.06 kW of electrical power (47,427.26 hp) to operate. The compressor was priced at \$2,100/kW, resulting in a total cost of \$74,113,326.53. All these results are given in Table 9 below:

Table 9: CO<sub>2</sub> Compressor Results

|                                       |                                                    |                                                               |
|---------------------------------------|----------------------------------------------------|---------------------------------------------------------------|
| Number of Stages   $r_{stage}$        | 4   2.74                                           | Stages   compression ratio/stage                              |
| $P_{CO_2,in} / P_{CO_2,out}$          | 2.1288   32.2851 / 120   1763.51                   | atm   psia / atm   psia                                       |
| CO <sub>2</sub> Capture Mass Flowrate | 105.69   3,335,409.15<br>838,844.37   3,676,654.86 | kg/s   metric tons/year<br>lb <sub>m</sub> /hr   US tons/year |
| Total Power Load                      | 35,292.06   47,427.26                              | kW   hp                                                       |
| Total Cost                            | \$74,113,326.53                                    | \$ for 1 four-stage compressor                                |

### 3.7 Cooling Water Results

Using Eq. 5, the heat demand (in kW thermal) was determined for each flow stream that requires cooling, including the flue gas entering the Absorber Column, the lean amine entering the Lean Amine Storage Tank, the CO<sub>2</sub> and water entering the pre-compressor condenser, and the inter-stage cooling of the CO<sub>2</sub> compressor. The largest contributor by far to the cooling demand is the Lean Amine Cooler, which requires a large amount of cooling water due to its large lean amine mass flow cycling through the Storage Tank. The lean amine must cool from the exit of the Rich/Lean Amine Heat Exchanger at 71.68°C down to 40°C before it can enter the Absorption Columns. Applying Eq. 5 to solve for the amount of cooling water flow results in the gpm flowrate required to achieve the desired cooling load for each demand. Using the existing cooling tower data for Blue Water, I derived an approximate cost and power load for running the cooling fans of the tower. All the data is summarized below in Table 10:

Table 10: Cooling Water Demand Results

|                                                |                            |                   |
|------------------------------------------------|----------------------------|-------------------|
| Flue Gas Pre-Cooler                            | 60,661.17   27,602.93      | kW thermal   gpm  |
| Lean Amine Cooler                              | 309,503.86   140,834.97    | kW thermal   gpm  |
| Regeneration Column Condenser                  | 10,738.05   4,886.19       | kW thermal   gpm  |
| CO <sub>2</sub> Compressor Inter-Stage Cooling | 28,275.03   12,804.85      | kW thermal   gpm  |
| Total Cooling Load                             | 409,178.11   186,128.94    | kW thermal   gpm  |
| Total Cooling Tower Cost, Power Load           | \$14,447,295.52   2,413.54 | \$ for Tower   kW |

### 3.8 Plant Footprint Results

Siting sufficient acreage for the capture plant is an important part of its installation. First, piping must be arranged to take the flue gas from each HRSG exhaust stack and redirect the flue to each Absorber Column, sited in the blue area. Then, the rich amine solvent mix flows across to the Amine Heat Exchanger and Regeneration Column, both sited in the green area. Finally, the capture CO<sub>2</sub> flows to the CO<sub>2</sub> Compressor and is prepared for transport. The compressor and all other auxiliary pumps and the cooling tower are located in the red area. The acreage of the capture plant totals 2.5<sup>+</sup> acres. All proposed areas are free of underground conduits and footers, as well as away from any overhead electrical lines. A visual overview of the existing NGCC plant and the proposed areas for the CCS plant are given below in Figure 2:



Figure 2: Aerial Overview of Blue Water Energy Center – Proposed Siting for CCS

## 4.0 Impact/Financial Benefits

### 4.1 Impact on Plant Operations

The most significant impact of adding carbon capture on the operation of Blue Water will be the steam turbine derate, which is due to the diversion of main steam away from the steam turbine and used instead for steam heating inside the capture system's Reboiler. To supply sufficient heating, an estimated 1,415.72 kpph (178.38 kg/s) of steam is required at full thermal load, which leaves 666.28 kpph for power generation. This amount of steam can only generate an estimated 141.77 MW of power in the steam turbine, resulting in an overall maximum derate of 301.23 MW, or 68% loss in maximum power output. This will significantly impact the maximum power output of the plant, reducing the total gross output of Blue Water from 1,127 MW to 825.77 MW.

Additionally, the capture equipment will also draw a substantial amount of power. The original, unmodified auxiliary power load – required for running all the existing plant equipment – is 2% of the gross output of the plant (which produces a maximum of 1,127 MW gross output), or 22.54 MW. The capture equipment will need 64.56 MW of power from all the added carbon processing equipment. This will further reduce the maximum gross output of the plant from 825.77 MW to 761.21 MW. The true maximum net generation of the plant will be 738.67 MW, or 22.54 MW less than the gross output. Table 11 below provides a summary of all the main CCS equipment power demand, all labeled in kW:

Table 11: Capture System Auxiliary Power Load

|                                    |              |
|------------------------------------|--------------|
| CO <sub>2</sub> Compressor         | 35,292.06 kW |
| Flue Gas Blowers                   | 16,574.42 kW |
| Lean Amine Cooler Water Pump       | 5,411.76 kW  |
| Cooling Tower Fans                 | 2,413.54 kW  |
| Lean Solvent Pump                  | 1,586.26 kW  |
| Rich Solvent Pump                  | 1,541.63 kW  |
| Flue Gas Pre-Cooler Water Pump     | 1,060.68 kW  |
| Compressor Intercooling Water Pump | 492.04 kW    |
| Regeneration Condenser Water Pump  | 187.76 kW    |
| Total:                             | 64,560.15 kW |

In addition to power draw, there will also be new water demand for cooling loads from the added capture system. The existing plant cooling tower has no additional capacity for cooling more water, thus a new, dedicated cooling tower will have to be built for the capture system. The new water demand is almost comparable to the existing plant cooling tower water demand, so the cost and power demand of the existing cooling tower was used as a basis for the cost and power load of the capture system's cooling tower. The existing cooling tower cost \$16,162,000 and could handle 208,220 gpm, requiring 2.7 MW of power to run the cooling fans. Comparatively, the total cooling load for the capture system is broken down into its component parts and summarized in Table 12 below:

Table 12: Capture System Cooling Load

|                                |                         |                             |
|--------------------------------|-------------------------|-----------------------------|
| Flue Gas Pre-Cooler            | 60,661.17   27,602.93   | kW <sub>thermal</sub>   gpm |
| Lean Amine Cooler              | 309,503.86   140,834.97 | kW <sub>thermal</sub>   gpm |
| Regeneration Column Condenser  | 10,738.05   4,886.19    | kW <sub>thermal</sub>   gpm |
| Compressor Inter-Stage Cooling | 28,275.03   12,804.85   | kW <sub>thermal</sub>   gpm |
| Total Cooling Load             | 409,178.11   186,128.94 | kW <sub>thermal</sub>   gpm |

The capture system's cooling tower will have to handle 89.39% of the gpm load that the existing cooling tower processes, which I used as a basis for the cost analysis and power load of the new cooling tower. The main consequence of adding a new cooling tower – besides cost and power draw – is that more water will need to be drawn from the St. Clair River at 16.67°C, and will discharge back to the river at 25°C. This is not expected to have any effect on the local wildlife or ecology of the river.

Lastly, during installation, there may be interruptions to service of the regular gas plant as pipes are fitted to connect with the existing steam system and HRSG stacks. It would be best to plan these interruptions in service during planned outage times for Blue Water, or at least during hours when demand is lowest on the grid. In this way, the plant could be part-loaded with only one of the two combustion gas turbines and HRSGs operational, while the other half gets modifications added. Additional project planning and scheduling would be needed to minimize the potential loss in the plant's power output and downtime.

## 4.2 Financial Impact of Adding Carbon Capture

There are multiple economic effects to DTE and Blue Water created by adding carbon capture to the existing natural gas combined cycle plant. First, the cost for the added capture equipment will be substantial. Table 13 below summarizes all of the costs listed from the results section:

Table 13: Total Cost of Carbon Capture Equipment

|                              |                   |
|------------------------------|-------------------|
| Absorber Columns             | \$ 145,667,797.84 |
| CO <sub>2</sub> Compressor   | \$ 74,113,326.53  |
| Regeneration Column          | \$ 24,486,604.33  |
| Cooling Tower                | \$ 14,447,295.52  |
| All Cooling Water Pumps      | \$ 11,443,586.19  |
| Amine Heat Exchanger         | \$ 6,730,931.68   |
| Lean Solvent Pump            | \$ 2,538,011.30   |
| Rich Solvent Pump            | \$ 2,466,604.52   |
| Reboiler Heat Exchanger      | \$ 1,847,737.41   |
| Flue Gas Blowers             | \$ 1,868,832.12   |
| Total Cost of All Equipment: | \$ 285,610,727.44 |

Additional, unlisted costs will be included for engineering design work, contracting and construction fees, pipes, valves, and insurance. All told, the cost of the project could reasonably exceed \$300 million.

There are also unseen costs, such as the lost revenue due to the decrease in total power output of the plant. These are due to new auxiliary capture equipment loads (discussed above in section 4.1) as well as lost power from the steam turbine due to the diversion of steam sent for steam heating in the solvent regeneration process in the Reboiler. The latter loss is far more significant, totaling up to \$259,034,633.07 per year if the plant runs at a 100% capacity factor. At the plant's typical capacity factor of 78.6933%, the expected loss is closer to \$160,410,680.30 at an average DTE charge rate of 16.31¢/kWh. Including the auxiliary power loads, the total profit loss from lost power is \$206,793,038.55.

Lastly, the LCOE of Blue Water will increase due to the capital cost of the added capture equipment and due to increased O&M costs for maintaining the capture equipment. Fuel costs, in this analysis, remain the same, as operationally DTE will need to decide if it will change the capacity factor of the plant to run more power more often in order to compensate for the loss in net power. The previous, unmodified plant LCOE was \$65/MWh (Chreston), or \$14/MWh in capital cost recovery, \$14/MWh is O&M costs, and \$37/MWh in fuel cost. Adding in \$285 million in new capital costs increases the capital cost LCOE by \$44.11/MWh, and a small amount of O&M change is expected as well. This leads to a new, CCS modified plant LCOE of \$110.49/MWh, which is very high for a gas plant.

### *4.3 Benefit of 45Q Tax Credits*

With the passing of the Bipartisan Budget Act of 2018 and the Inflation Reduction Act of 2022, Section 45Q of the tax code expanded the credit allowed for qualified facilities that capture carbon dioxide and use certain prevailing wage and apprenticeship requirements. These credits range from a base credit of \$12/metric ton of CO<sub>2</sub> for Enhanced Oil Recovery (EOR) and other qualified uses of CO<sub>2</sub>, to \$85/metric ton of CO<sub>2</sub> for geologically sequestered carbon captured by equipment installed by union trades that meet the wage and apprenticeship requirements. Given the capture rate and capacity factor for Blue Water – 2,624,743.32 metric tons of CO<sub>2</sub> per year – the range in tax credits will be between \$31,496,919.83 - \$223,103,182.14 per year, or \$377,963,037.97 - \$2,677,238,185.65 over the 12 years that Blue Water can qualify for 45Q credits.

Applying these credits to the above LCOE of the plant with carbon capture, the capital cost LCOE gets offset by \$34.46/MWh, leaving the final cost at \$23.65/MWh for capital expenditures. All other LCOE costs being equal as before, the new total modified LCOE becomes \$76.03/MWh. This is a reduction in LCOE from \$110.49/MWh down to \$76.03/MWh, which is still higher than the current unmodified LCOE of Blue Water at \$65/MWh.

## 5.0 Recommendations

The largest point of consequence for modifying Blue Water with carbon capture is the parasitic steam load required for desorbing the carbon dioxide from the MEA solvent in the Regeneration Column. The high demand for steam heating cuts into the steam used to generate power in the steam turbine at Blue Water, causing a significant loss in revenue from electricity sales. The greatest point of research following this report should be an investigation into limiting the amount of steam draw by the steam heating required for solvent desorption. If this project were to be attempted by DTE, I would recommend that a professional mechanical or chemical engineer review the steam system at Blue Water to determine a means of lowering the steam demand. This might include an investigation of additional means of steam generation that do not require more fossil fuel use – such as using alternative non-emitting fuels like hydrogen – or otherwise finding a more efficient means of achieving heat transfer from transferring heat from other points in the plant for solvent regeneration.

In order to minimize the steam losses to steam heating and solvent regeneration, I recommend taking the required 178.38 kg/s (1,415.72 kpph) of steam and reducing the pressure significantly down to 2.685 atm (39.46 psi) where the saturation temperature is 130.219°C, such that a  $> 10^{\circ}\text{C}$  approach can be achieved to heat the 120°C solvent in the Reboiler. At lower pressure, the temperature change from 600°C main steam down to 130.219°C saturation allows for greater heat transfer than the existing main pressure steam at 180 bar (2,610 psi) which has a saturation temperature of 356.992°C. Additionally, lower pressures have a greater change in enthalpy during vaporization/condensation, which extracts further energy from the steam. Specifically, the enthalpy  $h_{fg}$  increases from 777.749 kJ/kg at a pressure of 180 bar to 2,173.06 kJ/kg at a pressure of 2.685 atm. Overall, running the steam heating at lower pressure allows for greater heat transfer and minimizes steam diversion away from the steam turbine where it is more desirably used for generating power. Reducing the pressure can be done a number of ways, but the least invasive strategy would be to add a splitting valve to the main steam header (high pressure steam) coming from the HRSGs and expand the pipe as it moves towards the Reboiler. This pipe expansion will decrease the pressure of the main steam down to 2.685 atm. The other – more intrusive – method would be to physically cut into the steam turbine or IP-LP cross over pipe and add an extraction pipe with a limiting valve to split off steam at 2.685 atm of pressure and send the steam to the Regeneration Column and Reboiler. This method would have less overall control of the mass flow of steam, and could have detrimental consequences on the typical operation of the steam turbine. It would also likely be more expensive to add such an invasive modification to such an expensive piece of equipment.

Lastly, in the course of the economic analysis for adding carbon capture to Blue Water, I discovered that continuing the plant's average capacity factor (78.6933%) for operation, the tax credits gained over the profit loss from power derate equated to only \$16,310,161.14 per year, which over the 12 years of eligibility for receiving tax credits would only result in a profit of \$195,721,933.71. This value falls short of the \$285,610,727.44 price tag for adding the capture equipment by a shortfall of \$89,888,793.73. In order to break even, the plant will need to run slightly less often over the course of the year by 3.24% less capacity factor. This means that adjusting power output to run at a maximum capacity factor of 75.4539% will allow for Blue Water to make enough money to cover the cost of the carbon capture system and its equipment. It is important to DTE to make sure this project is profitable.

The other consideration is what to do with the carbon dioxide once it has been captured. While it is outside the scope of this report to recommend a definitive destination for the CO<sub>2</sub>, it is important to mention that the current 12 year tax credit collection period is finite and will not last for the entire life of the plant till 2052. Creative solutions could be found as the market for carbon expands over the next decade, which might include industrial uses beyond the current uses in enhanced oil recovery or cement making. My recommendation is that DTE fully explores its options for post-tax credit revenue for the captured carbon before embarking on implementing carbon capture at Blue Water.

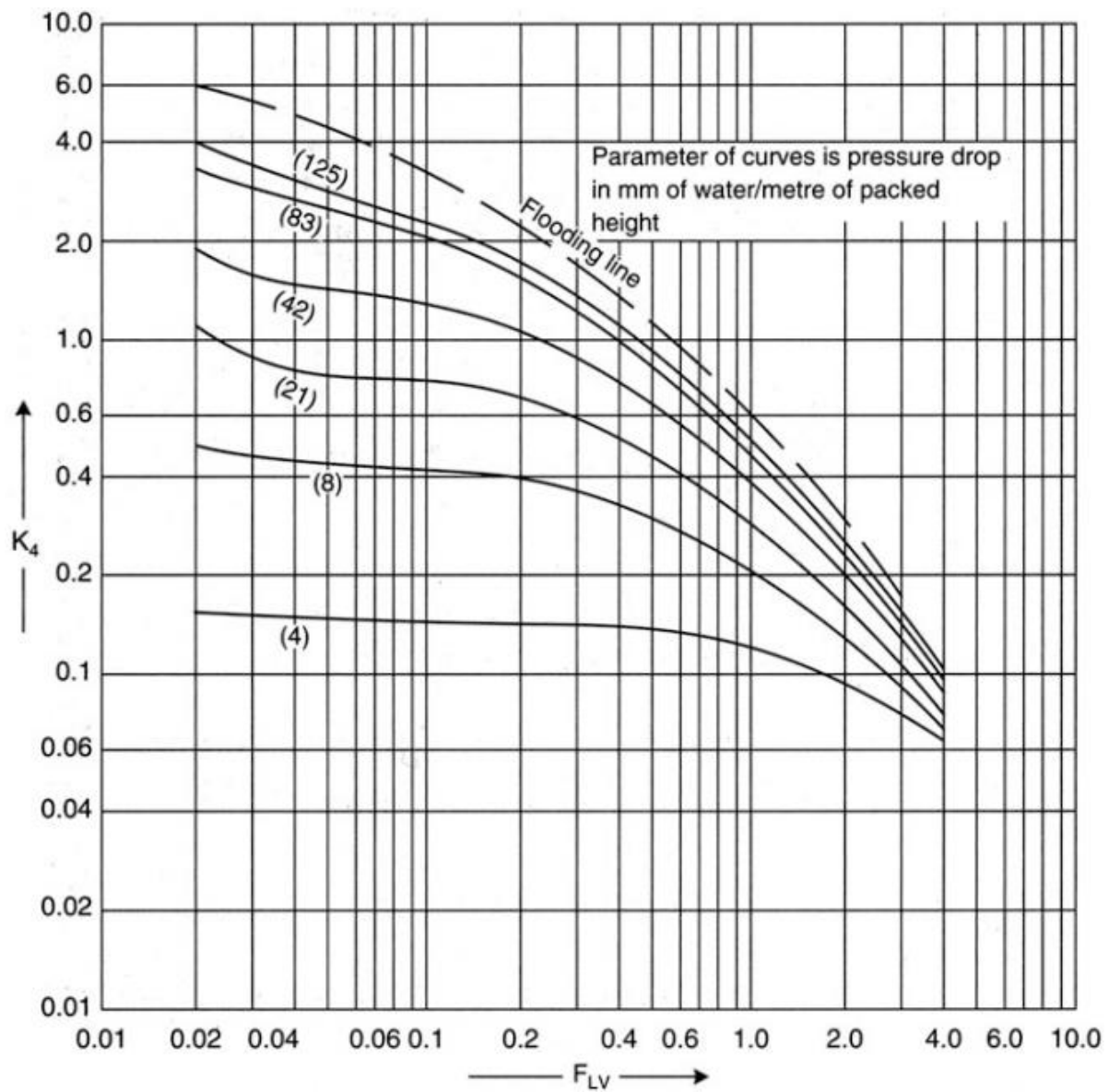
## 6.0 Summary/Conclusions

Implementing carbon capture on an existing natural gas combined cycle plant is no small task. There are numerous details that collectively affect the operational and economic outcomes for the plant and the power producer's bottom line. For DTE, installing carbon capture at Blue Water Energy Center is a near breakeven venture, given the main constraint that the steam demand for the solvent regeneration process constitutes 68% of the total steam generation at Blue Water. This means only 32% of the total steam generation can be used towards power production and revenue service, limiting the plant to a maximum net output of 738.67 MW. The gross generation is limited to 825.77 MW with 388.33 MW of total losses, 301.23 MW of which is from the steam diversion for use in releasing CO<sub>2</sub> and regenerating the MEA amine solvent.

The capture plant is expected to cost \$285,610,727.44, which will be able to be covered by the tax credit incentives from 45Q of the tax code. Adding CCS can capture up to 3,335,409.15 metric tons of CO<sub>2</sub> per year at 100% output, though at the existing capacity factor of 78.6933% that figure drops to 2,624,743.32 metric tons. At a maximum tax benefit of \$85 per ton of CO<sub>2</sub> sequestered, this means the plant can expect \$223,103,182.14 of tax relief per year, or a total program tax benefit of \$2,677,238,285.65 over 12 years. In order to even out the cost of building the capture plant and the lost revenue from the derate of the plant with the tax credits, Blue Water must run at a lower capacity factor of 75.4539%.

The analysis demonstrates that, while not optimally economically feasible, a business case can be achieved that will support a positive economic incentive to implement carbon capture at Blue Water. In the near future, technological improvements in carbon capture will pave the way to lower costs and less steam heating demand which will make carbon capture attractive for power producers. Environmental incentives need to be considered when deciding to build a capture plant, as it is imperative that we as a global community begin to make strides in reducing our carbon footprint to reduce the increase in global warming. It is important that companies like DTE invest in this technology and continue to improve on its implementation in order to drive innovation and efficacy. Without carbon capture, we have no means of mitigating our emissions and ensuring a safe, carbon neutral environment for our future. We must act on climate change, and decarbonizing our power sector is an excellent place to start.

## 7.0 Appendices



Appendix A: Generalized Pressure Drop Correlation

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